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# ***NumPY Library***

First import Numpy library

Import numpy as np

To find numpy version:

print(np.__version__)

To convert list into array use function:

arr=np.array([1,2,3])

print(arr)

To find dimension of array use the function:

Print (array name.ndim)

To find type of array use the function:

type(name of array)

## ***Numpy ONES functions***

onr=np.ones(5)

print(onr)

#It only print float value To find data

type of array use the function:

functionName.dtype

Make matrix using ONES function : onr1=np.ones((x,y))

# x=number of rows,y=number of coloums

print(onr1)

To change the data type into INTEGER of ONES function use the method:

onr2=np.ones((3,5), dtype=int) print(onr2)

To change the data type into BOOL of ONES function use the method:

onr2=np.ones((3,5), dtype=bool) print(onr2)

To change the data type into STRING of ONES function use the method:

onr2=np.ones((3,5), dtype=str)

print(onr2)

***Numpy ZEROS functions***

onrz=np.zeros((4,5));

print(onrz) # It also give float value

To find data type of ZEROS function :

FunctionName.dtype

To change the data type into INTEGER of ZEROS function use the method: onr2=np.ones((3,5), dtype=int)

print(onr2)

To change the data type into BOOL of ZEROS function use the method: onr2=np.ones((3,5), dtype=bool) print(onr2)

To change the data type into STRING of ZEROS function use the method:

```
onr2=np.ones((3,5), dtype=str)
print(onr2)
```

To make an matrix with random number use the

function: emp=np.empty((3,4)) print(emp)

To make an array of some sequence numbers use the

method : arg=np.arange(1,13)

print(arg)

#In this function 1st number is included and last number is excluded

To give some increament use the

function: arge=np.arange(0,13,2)

print(arge)

When you have to make some equal numbers between two numbers use LINESPACE function:

ls=np.linspace(1,10,20)

print(ls)

#It make 20 numbers b/w 1 and 10 having equal distance

## ***SHAPING an Array***

Convert 1D array to 2D array use function:

arrng2=arg.reshape(4,3)

print(arrng2) #where 4 are rows and 3 are coloums Convert 1D array to 3D array use function: Arrng3=arg.reshape(4,3,2) print(arrng2) #multiplication of attributes =total attributes in 1D array

For Converting 2D or 3D Array to 1D form these functions are used: Array name.ravel()

Array name.flatten()

For tranpose(Converting Row into Coloums and vice versa) of a matrix use the function:

Array name.transpose()

# ***Do Math Functions using NUMPY***

To add two arrays use this function:

Arr1+Arr2 or

Np.add(arr1,arr2)

To subtract two arrays use this function:

Arr1 – Arr2 or

Np.subtract(arr1,arr2)

To multiply two arrays use this function:

Arr1*Arr2 or

Np.multiply(arr1,arr2)

To divide two arrays use this function:

Arr1 / Arr2 or

Np.divide(arr,arr2)

For DOT multiply of two arrays use this function:

Arr1@Arr2 # it multiply number of rows into number of coloumn or

Np.dot(arr1,arr2)

To find maximum number in array use this function:

Array name.max

To find maximum number in row in array use this function:

Array name.max(axis=0)

To find maximum number in colomn in array use this function:

Array name.max(axis=1)

To find minimum number in array use this function:

Array name.min

To find minimum number in row in array use this function:

Array name.min(axis=0)

To find minimum number in colomn in array use this function:

Array name.min(axis=1)

To find that index at which maximum number lies use this function:

Np.argmax(array name)
# remember indexing starts from 0

To find that index at which minimum number lies use this function:

Np.argmin(array name)

To find square root of an array use this function:

Np.sqrt(array name)

To find the mean value of an array use this function:

Np.mean(array name)

To find standered daviation of an array use this function:

Np.std(array name)

To find the log of an array use this function:

Np.log(array name)

To find the log10 of an array use this function:

Np.log10(array name)

To find exponential value of an array use this function:

Np.exp(array name)

```
arr=np.arange(1,101).reshape(10,10)
print(arr)
```

To find any index in an array use this function:

Array name[Row point,Coloum point]

To show the whole row use this function:

Array name[index of row]

To show the whole coloum use this function:

Array name[:,index of coloum]

To print the same array also use these functions: arr[:] arr[:,] arr[::]

To pick some specific rows and coloms use this function:

Array name[2:5,6:8]

To find memory size use this function:

Array name.itemsize # it gives 4bytes

Proof::

Array name.dtype # it give 32

32/8=4bytes # 8bits=1byte

Take two arrays and print them:

arr1=np.arange(1,17).reshape(4,4)

arr2=np.arange(17,33).reshape(4,4)

Also take two list and print them also :

list1=[1,2,3,4]

list2=[5,6,7,8]

list1+list2

```
[1, 2, 3, 4, 5, 6, 7, 8]
```

When lists are added it increases the indexes while when we add arrays the result was opposite:

```
array([[18, 20, 22, 24],
       [34, 36, 38, 40],
[26, 28, 30, 32],
       [42, 44, 46, 48]])
```

For addition of indexes in arrays like lists use this function:

np.concatenate((arr1,arr2)) or use
np.vstack((arr1,arr2))

# for rows increase

np.concatenate((arr1,arr2),axis=1) or use

np.hstack((arr1,arr2))  # for coloum increase

## ***Methods of String using NumPy***

To Concatinate two Strings use the method :

Np.char.add(String1,String2)

To change a String into Lower case use the method:

Np.char.lower(String name)

To change a String into Upper case use the method:

Np.char.upper(String name)

To write a String between characters at center use the method :

np.char.center(sname,80)

If you want to avoid whitespaces use the method :
np.char.center(sname,80, fillchar="*")

For splitting a String to seprate word use the method:

np.char.split(sname)

For splitting a String to lines use the method:

np.char.splitlines("Dev\nValley\nSoftware\nHouse")
For character shaping use the method:

```
name2="aSamad"

name3="Daska"

np.char.join(["/",":"],[name2,name3])
```

For replacing a word use the method:

```
np.char.replace(sname,"Software","s/
w")
```

To check that two or more Strings are equal use the method:

```
np.char.equal(name2,name3)
```

To check number of character in a String use the method: `np.char.count(name2,"a")`

To find the index of word in a String use the method :

```
np.char.find(sname,"Dev")
```